

Plants vs Zombies 2 - UML Package and Class Design

Advanced-Programming-Project-Plants-VS-Zombies-2



Group name: Entropic

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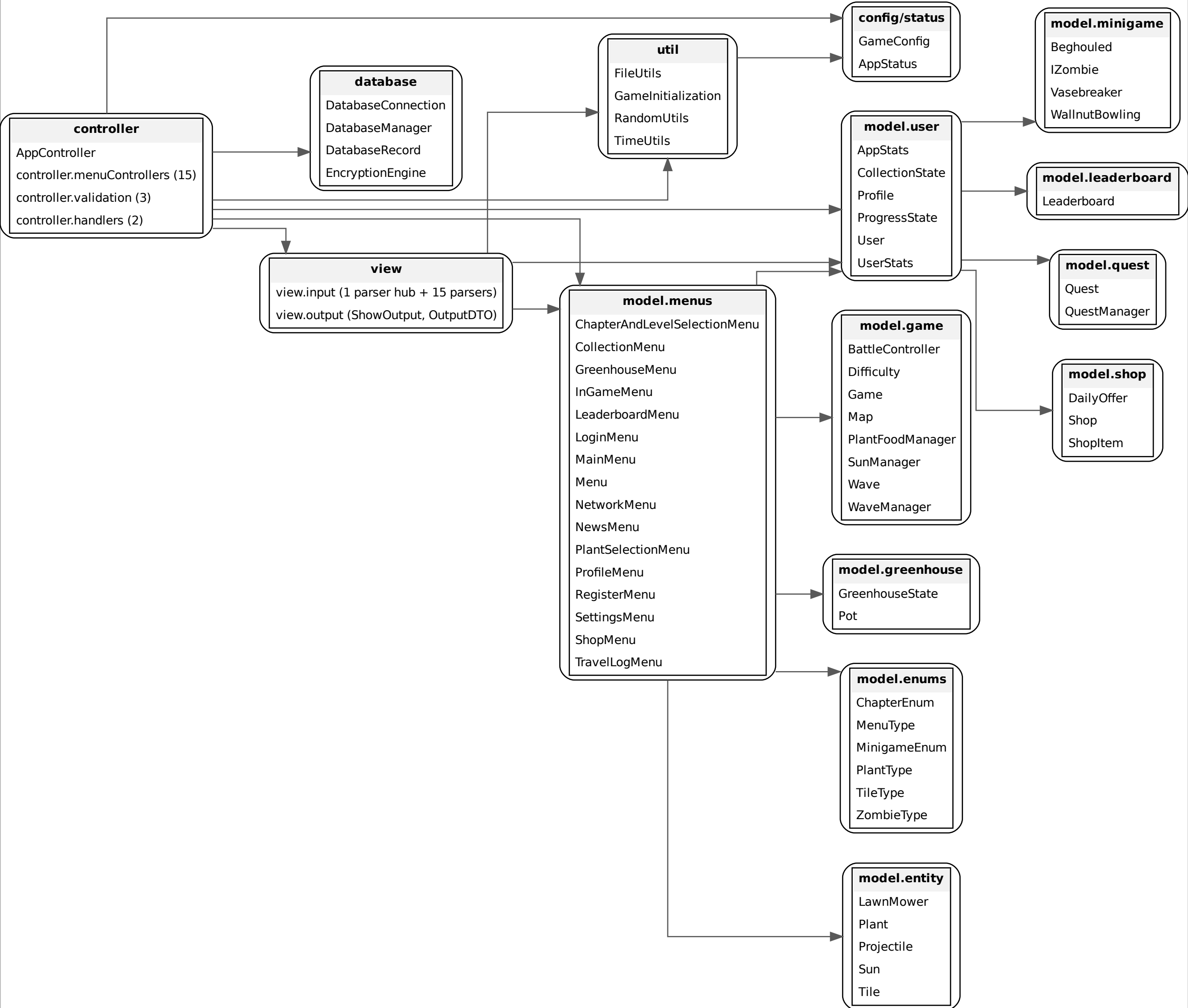
Group Members:

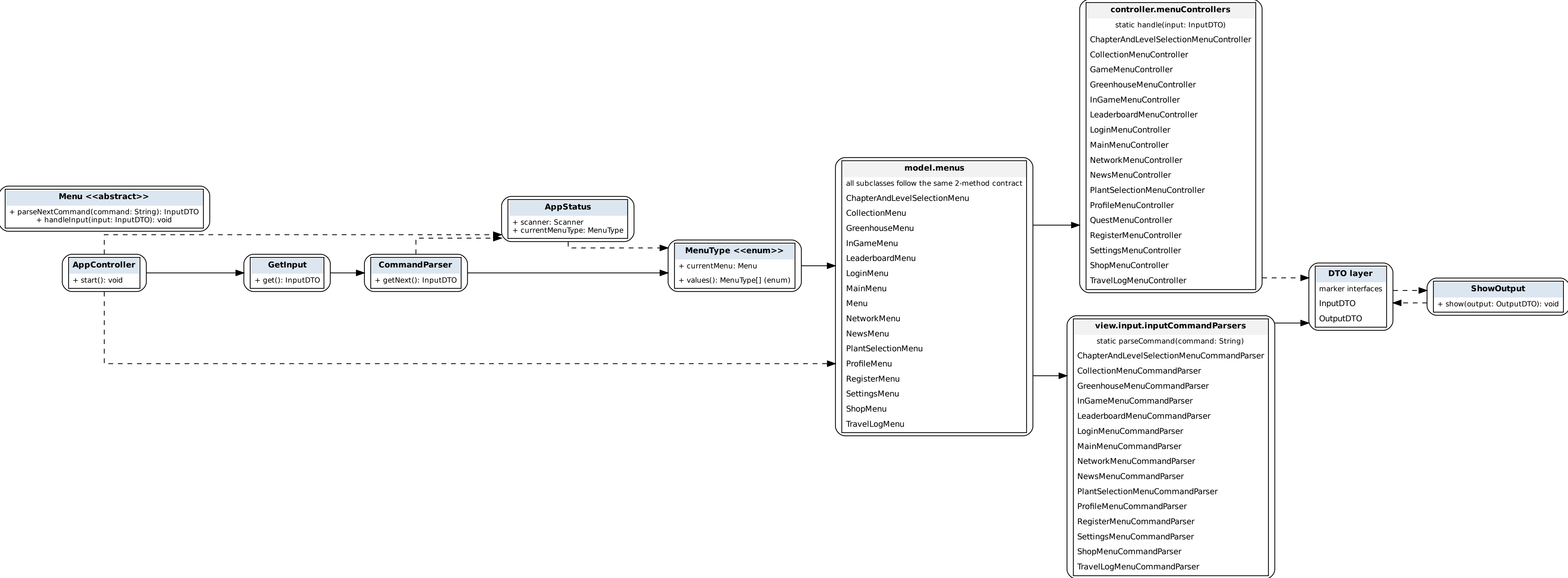
Ali Ariakia

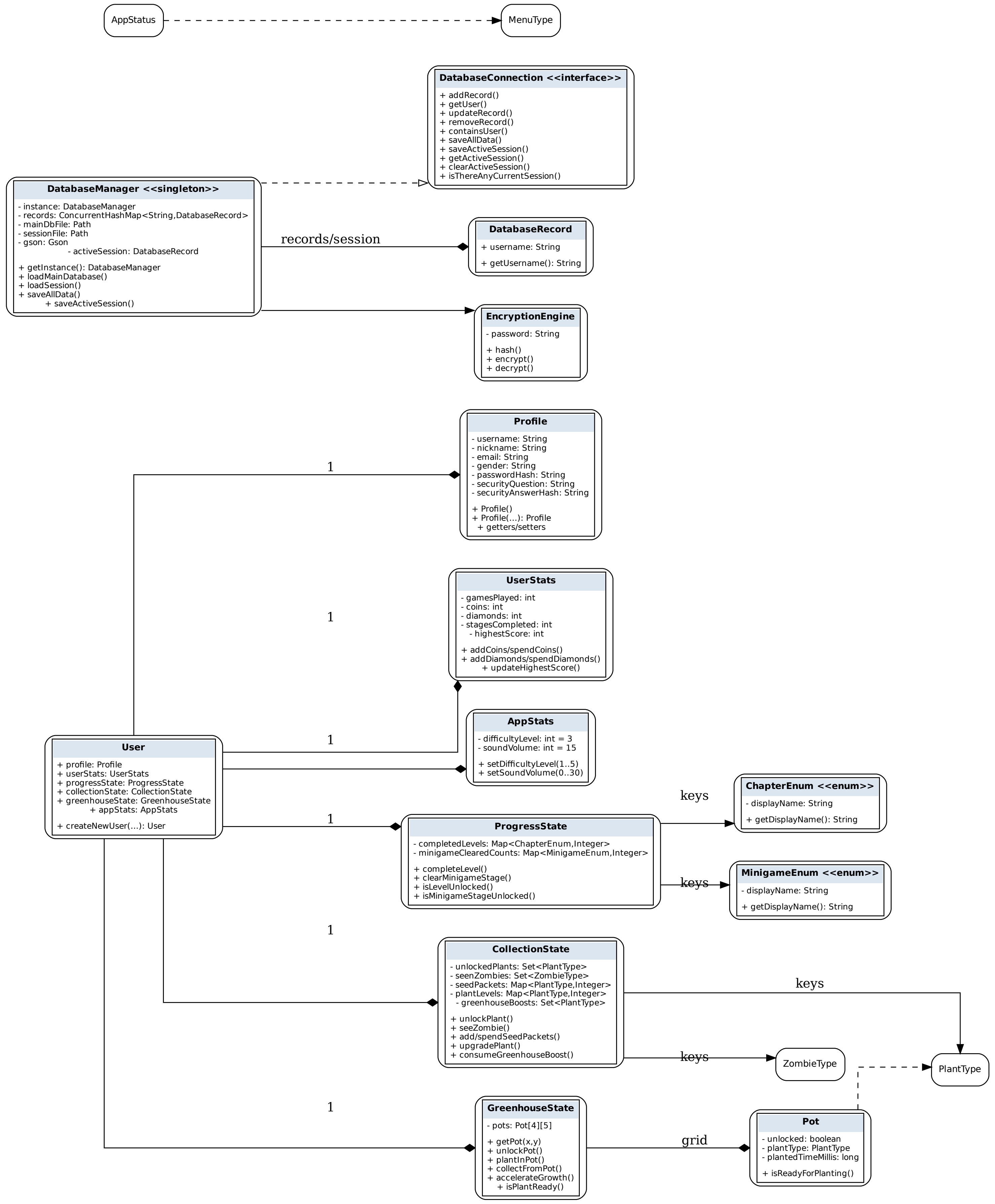
Fatemeh Mostafavi

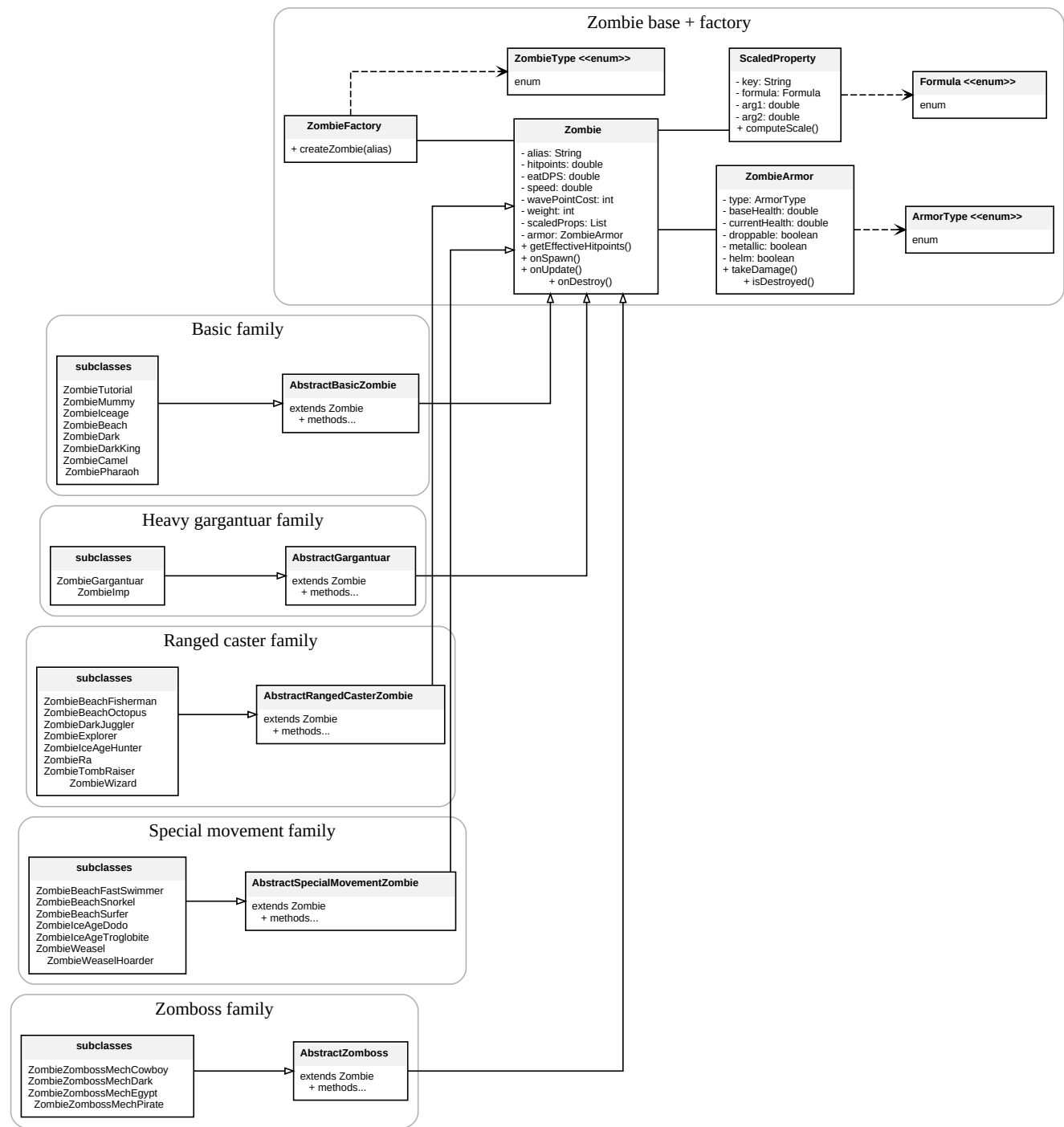
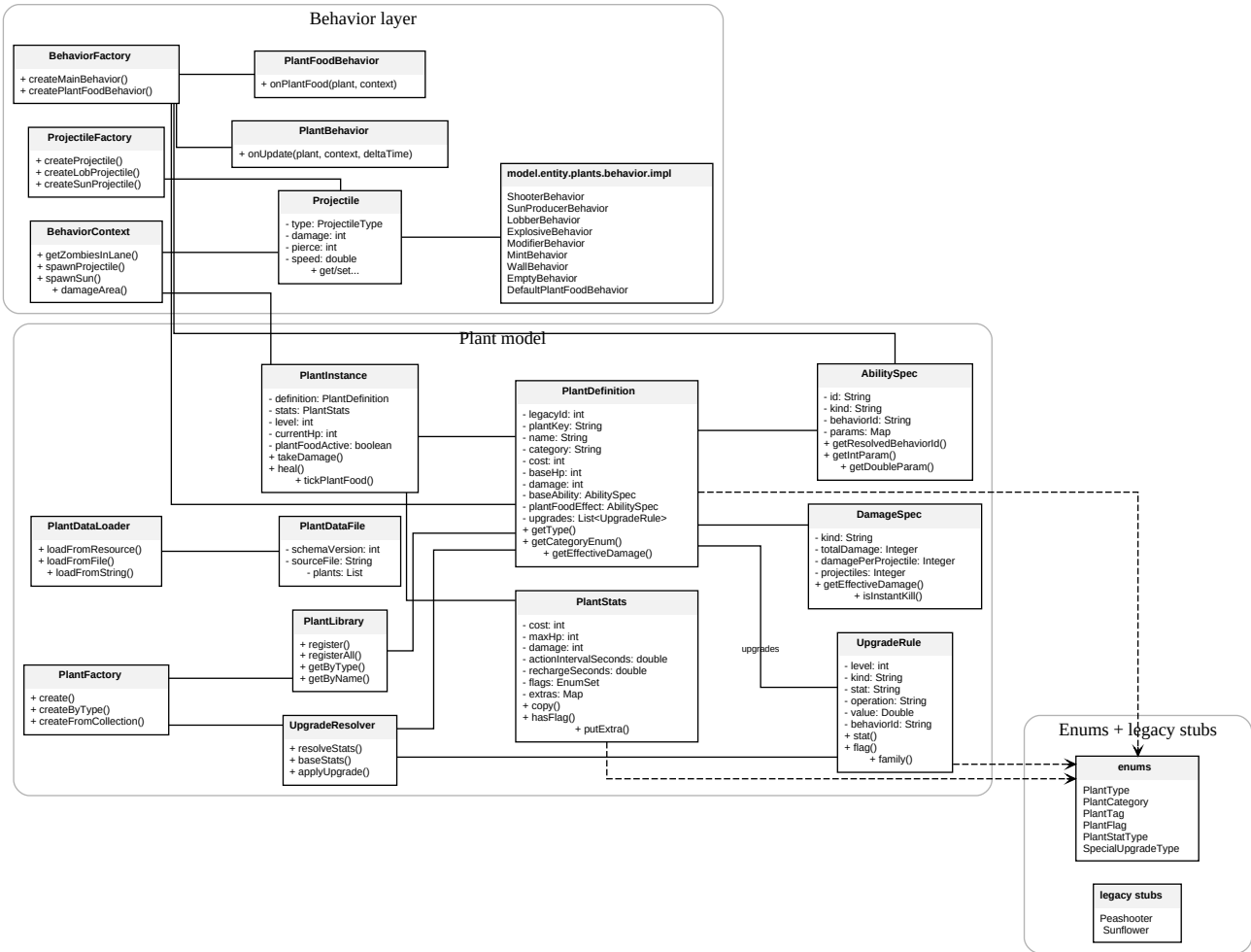
Mehdi HajEbrahimi

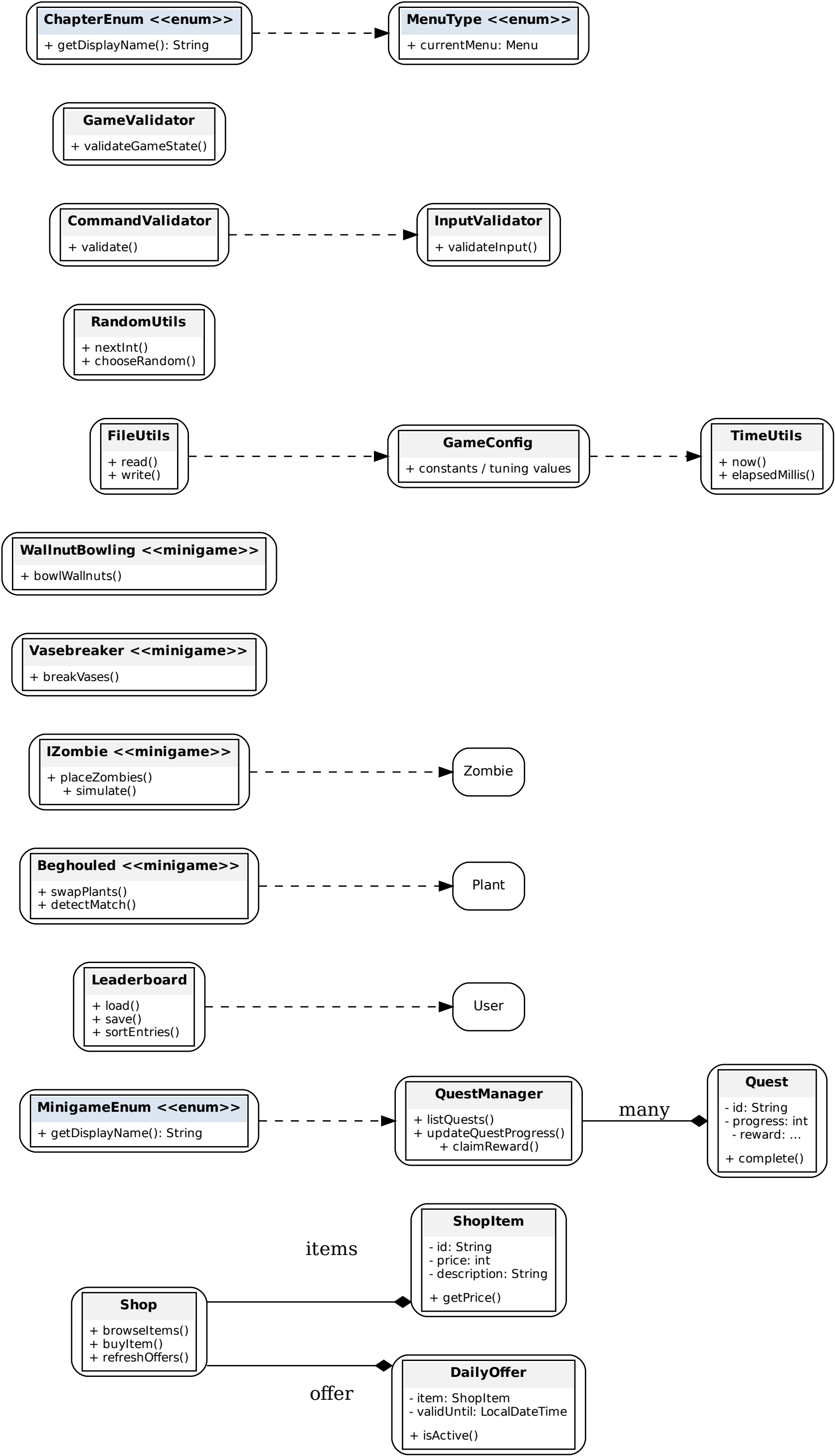
Package overview

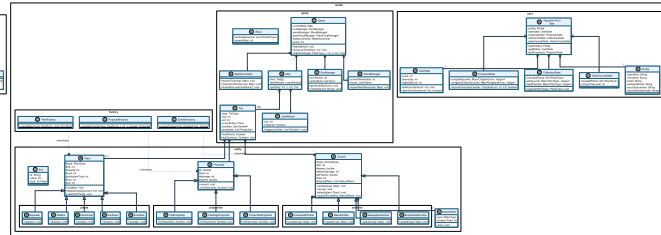
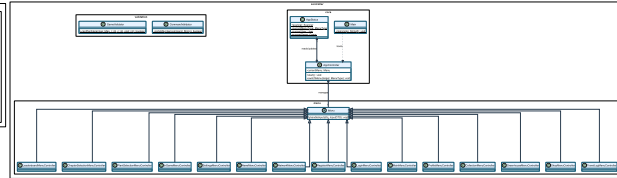
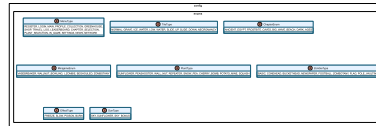












All in One

Class Catalog

(default package)

Main — model/controller class

config

GameConfig — global config placeholder [skeleton / inferred]

controller

AppController — main event loop and input dispatch

controller.menuControllers

ChapterAndLevelSelectionMenuController — static handler for its menu

CollectionMenuController — static handler for its menu

GameMenuController — static handler for its menu

GreenhouseMenuController — static handler for its menu

InGameMenuController — static handler for its menu

LeaderboardMenuController — static handler for its menu

LoginMenuController — static handler for its menu

MainMenuController — static handler for its menu

NetworkMenuController — static handler for its menu

NewsMenuController — static handler for its menu

PlantSelectionMenuController — static handler for its menu

ProfileMenuController — static handler for its menu

QuestMenuController — static handler for its menu

RegisterMenuController — static handler for its menu

SettingsMenuController — static handler for its menu

ShopMenuController — static handler for its menu

TravelLogMenuController — static handler for its menu

controller.validation

CommandValidator — design stub

GameValidator — design stub

InputValidator — design stub

database

DatabaseConnection — persistence interface

DatabaseManager — singleton persistence manager ; implements DatabaseConnection

DatabaseRecord — serializable user/session record

EncryptionEngine — hash/encryption helper

model.enums

ChapterEnum — enumeration

MenuType — type enumeration

MinigameEnum — enumeration

PlantType — type enumeration

TileType — type enumeration

ZombieType — type enumeration

model.entity

LawnMower — lane fail-safe / mower entity

Plant — abstract plant entity

Projectile — projectile entity

Sun — sun collectible entity

Tile — one board cell

model.entity.plants

Peashooter — design stub

Sunflower — design stub

model.entity.zombies

BasicZombie — design stub

ConeheadZombie — design stub

model.game

BattleController — battle orchestration service

Difficulty — difficulty abstraction

Game — battle session / match state

Map — board grid and tile management

PlantFoodManager — plant-food inventory logic

SunManager — sun generation and collection logic

Wave — one zombie wave definition

WaveManager — wave spawning and progression logic

model.greenhouse

GreenhouseState — greenhouse pot grid state

Pot — one greenhouse slot

model.leaderboard

Leaderboard — high-score ranking

model.menus

ChapterAndLevelSelectionMenu — menu screen / command dispatcher ; extends Menu

CollectionMenu — menu screen / command dispatcher ; extends Menu

GreenhouseMenu — menu screen / command dispatcher ; extends Menu

InGameMenu — menu screen / command dispatcher ; extends Menu

LeaderboardMenu — menu screen / command dispatcher ; extends Menu

LoginMenu — menu screen / command dispatcher ; extends Menu

MainMenu — menu screen / command dispatcher ; extends Menu

Menu — abstract menu contract

NetworkMenu — menu screen / command dispatcher ; extends Menu

NewsMenu — menu screen / command dispatcher ; extends Menu

PlantSelectionMenu — menu screen / command dispatcher ; extends Menu

ProfileMenu — menu screen / command dispatcher ; extends Menu

RegisterMenu — menu screen / command dispatcher ; extends Menu

SettingsMenu — menu screen / command dispatcher ; extends Menu

ShopMenu — menu screen / command dispatcher ; extends Menu **TravelLogMenu** — menu screen / command dispatcher ; extends Menu

model.minigame

Beghouled — minigame mode

IZombie — minigame mode

Vasebreaker — minigame mode

WallnutBowling — minigame mode

model.quest

Quest — one quest definition

QuestManager — quest tracking and rewards

model.shop

DailyOffer — time-bound shop offer

Shop — shop inventory and purchase flow [skeleton / inferred]

ShopItem — shop item model [skeleton / inferred]

model.status

AppStatus — global runtime state [skeleton / inferred]

model.user

AppStats — app settings and difficulty

CollectionState — unlocks, cards, and collection progress

Profile — authentication and identity data

ProgressState — chapter/minigame progression

User — aggregate root for player state

UserStats — persistent player statistics

util

GameInitialization — bootstraps initial ap

FileUtils — file helpers [skeleton / inferred]

RandomUtils — random helpers [skeleton / inferred]

TimeUtils — time helpers [skeleton / inferred]

view.input

CommandParser — parses raw CLI command into InputDTO

GetInput — input facade

[InputDTO — DTO marker interface [skeleton / inferred

view.input.inputCommandParsers

ChapterAndLevelSelectionMenuCommandParser — parses raw CLI command into InputDTO

CollectionMenuCommandParser — parses raw CLI command into InputDTO

GreenhouseMenuCommandParser — parses raw CLI command into InputDTO

InGameMenuCommandParser — parses raw CLI command into InputDTO

LeaderboardMenuCommandParser — parses raw CLI command into InputDTO

LoginMenuCommandParser — parses raw CLI command into InputDTO

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ShopMenuCommandParser — parses raw CLI command into InputDTO

TravelLogMenuCommandParser — parses raw CLI command into InputDTO

view.output

OutputDTO — DTO marker interface

ShowOutput — output facade / renderer hook